

Mobile Computing - Syllabus

UNIT - I

Introduction – Applications – A short history of wireless communication.

Wireless transmission: Frequencies for radio transmission – Signals – Antennas – Signal propagation – Multiplexing – Modulation –

Amplitude – Shift Keying – Frequency Shift Keying – Phase Shift Keying – Spread spectrum.

UNIT - II

SDMA – FDMA – TDMA – Fixed TDM – Classical Aloha – CDMA – Cellular Systems for Mobile communication –

GPRS – Satellite Systems – Basics applications – Broadcast Systems – Digital Audio Broadcasting – Digital Video Broadcasting –

Local development of application in mobile computing platform.

UNIT - III

Infrared vs Radio Transmission – Infrastructure Networks – Ad hoc Networks – IEEE802.11 –

System Architecture – Protocol – Product – Bluetooth – User scenarios – Bluetooth Architecture –

Introduction to wireless LAN services – Location Reference Model.

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UNIT - IV

Mobile IP – Goals – Assumptions – Entities and terminology – IP Packet delivery – Agent advertisement and discovery – Registration – Tunneling and encapsulation – Optimizations – Dynamic Host Configuration Protocol (DHCP) –
Routing – DSR – AODV – Alternative Mechanisms.

UNIT - V

Introduction – Protocol Architecture – Wireless Markup Language (WML) – WML Script –
Applications – Wireless Technology Application (WTA) – Wireless Technology Application Architecture (WTAA).